Jack Pham

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EDUCATION

University of California at Berkeley **Berkeley, CA**
Bachelor of Science in Electrical Engineering and Computer Sciences, GPA: 3.33 *August 2025 – December 2027*

Cypress College **Cypress, CA**
Associate Degree for Transfer in Mathematics and Physics, GPA: 3.94 *February 2023 – May 2025*

TECHNICAL SKILLS

Programming & Data Analysis: Python, C, C++, Java, SQL, MATLAB, Bash, NumPy, SciPy, cvxpy

Electrical Engineering & Embedded Systems: Embedded C/C++, circuit design, PCB layout, Arduino, ESP32, ROS, Gazebo, sensors and actuation, signal processing fundamentals, soldering
Engineering Tools & Platforms: Git, Linux, VS Code, Jupyter, Docker, AutoCAD

RESEARCH EXPERIENCE

Texas Advanced Computing Center (TACC), UT Austin **Austin, TX**
Summer Researcher *June 2024 – August 2024*

- Developed a multi-robot task allocation and path-planning framework using Python, cvxpy, and ROS, improving simulated disaster-response efficiency by 70%.
- Designed, simulated, and validated multi-agent coordination algorithms in Gazebo, leveraging TACC's supercomputing cluster to accelerate large-scale optimization experiments.
- Implemented benchmarking tools that reduced planning computation time for robotic coordination cycles by 45%.

WORK EXPERIENCE

North Orange County Community College District **Cypress, CA**
Physics Tutor *February 2025 – July 2025*

- Developed Python and Jupyter simulations to visualize concepts in electromagnetism, relativity, and thermodynamics across 40+ tutoring hours.
- Supported physics laboratory sessions by explaining experimental setups, measurement tools, and data collection methods using multimeters and electrical instrumentation.
- Helped students troubleshoot lab experiments by diagnosing wiring errors, measurement inconsistencies, and conceptual misunderstandings in mechanics, circuits, and electromagnetism.

Charger Experience at Cypress College **Cypress, CA**
Student Ambassador *January 2024 – December 2024*

- Automated parts of student data processing during outreach events using Python scripts and SQL queries and built live-updating web visualizations to analyze student engagement trends.
- Processed and resolved 75+ student cases per semester by coordinating with admissions and student services, improving response efficiency by 30%.
- Represented the college at 12 outreach events, engaging over 500 prospective students and organizing campus tours and onboarding sessions.

UNIVERSITY ACTIVITIES

Underwater Robotics Team, UC Berkeley **Berkeley, CA**
Electrical Subteam Member

- Investigating power dissipation and thermal behavior of NVIDIA Jetson modules in autonomous underwater robots.
- Assisting with electrical subsystem integration and debugging systems connected to 16 sensors and actuators.

PROJECTS

AI Kitchen Accessibility Assistant | ESP32-CAM, Computer Vision, Speech Recognition **Spring 2026**

- Built embedded vision system recognizing kitchen tools and responding to voice commands for accessibility applications.
- Trained object recognition models on university HPC resources and deployed inference on ESP32-CAM hardware.

EECS16A Acoustic Positioning System | Python, Signal Processing **UC Berkeley, Fall 2025**

- Built embedded vision system recognizing kitchen tools and responding to voice commands for accessibility applications.
- Validated localization using simulated and real microphone recordings under noisy conditions.

AWARDS AND PRESENTATIONS

- *Kathy Godshalk Memorial Scholarship for Science — 2025*
- Presented at Southern California Conferences for Undergraduate Research, San Bernardino, CA, November 2024
- Presented at STARS Celebration Conference, San Diego, CA, September 2024